Methods for agent-based computer simulations

Version 1

In the agent-based model, each cell within the aggregate is represented by a point corresponding to its centre of mass. Cells undergo proliferation and differentiate according to the 3-state differentiation model. The collective dynamics of the system is governed by the equation of motion of a single cell (see Appendix).

Cell movement arises from forces exerted by neighbouring cells. Following the formulation in [1], each cell experiences both passive and active interaction forces. The passive contribution includes differential adhesion, modelled as an attractive cell–cell interaction, and volume exclusion, which introduces short-range repulsion. The active contribution accounts for contractile protrusions formed between cell pairs, generating fluctuations that promote cell rearrangements and capture tissue fluidity. Similarly to the adhesion force strength, the protrusion force activation and deactivation rates are assigned in a cell-type specific manner, making the interaction preferential. Additionally, cell–cell friction is accounted for, such that the velocity of each cell depends not only on the net force acting on it but also on the velocities of its nearest neighbours. A detailed description of these forces is provided in the Appendix.

To study symmetry breaking, we define the segregation score parameter, which measures how many of a cell's neighbours share the same fate. The segregation scores of each cell are then averaged over all cells with a given fate, giving us the average segregation score or sorting index for each cell fate and allowing us to quantify the spatial segregation of different fates are over time. Additionally, in order to study polarisation, we track the distance between the centre of mass of the largest T+ (B-type) cluster and that of the entire aggregate over time. We use a threshold value of at least 20% the initial (24hpa) aggregate size for considering the largest T+ cluster; and the relative distance is normalised by the aggregate size at 25hpa (which is prior to the onset of patterning) providing a quantitative estimate of the degree of polarisation.

Simulations were run up to 75 hpa, matching the time window of the *in-vitro* experiments. Starting from an initial aggregate size of 600 cells at 24 hpa, we simulated cases with initial fractions of T+ cells of 0%, 20%, 50%, and 80%. To examine the effect of the initial aggregate size, additional simulations were performed with 300, 600, and 900 cells at 24 hpa for $\phi = 0\%$. For each condition, 10 independent simulations were run to ensure statistical robustness.

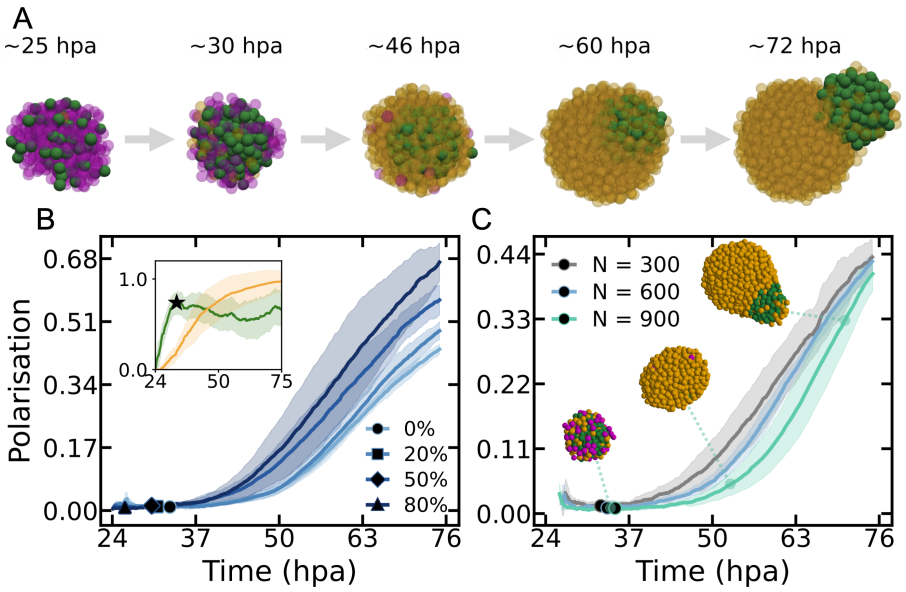


Figure 1: (A) Representative snapshots of a 3D agent-based simulation of gastruloid symmetry-breaking considering cell-cell mechanical interactions and cell fate transitions. Cells adhere to each other depending on their cellular fate (A, B or C) and send contractile cellular protrusions driving cell-cell rearrangements. (B) Polarisation vs time for different initial starting fractions ϕ and initial aggregate size N = 600. Polarisation is measured as the normalised center-of-mass distance between the B-cluster and the entire aggregate. The normalisation is done with respect to the intial size of the aggregate (see Material and Methods). Inset: Sorting index vs time for the B (green) and C (orange) clusters. The timepoint when the sorting index is maximum is indicated with black symbols. (C) Polarisation vs time for different initial number of cells at 24h for an initial starting fraction of $\phi = 0$. Larger aggregates polarise slower than the smaller ones. For each condition, mean $\pm$ SD is shown for 10 different simulations. For visualisation purposes, we consider cells with fewer than 4 neighbours within a cutoff distance of roughly 2 times the average T+ cell radius (see Appendix) to be delaminated, and are therefore not represented in the simulation snapshot images provided in panel (A) and the inset of panel (C).

References

[1] David Oriola, Miquel Marin-Riera, Kerim Anlaş, Nicola Gritti, Marina Sanaki-Matsumiya, Germaine Aalderink, Miki Ebisuya, James Sharpe, and Vikas Trivedi. Arrested coalescence of multicellular aggregates. *Soft Matter*, 18(19):3771–3780, 2022.